

Question 1 (15 marks)

- a) Write down the exact value of: 2
- i. $\sin 150^\circ$
 ii. $\operatorname{cosec} 315^\circ$
- b) Find the exact value of x if: 3
 $3 \log_{10} 4 - 2 \log_{10} x = 6$
- c) Shade the region(s) in the Cartesian plane which represent the simultaneous solution of $y \leq x + 4$ and $y \geq x^2 - 4$. 2
- d) Given that $\frac{\sqrt{2}+1}{\sqrt{2}-1}$ can be written in the form $p + q\sqrt{2}$, find the values of p and q . 2
- e) For what values of k is the expression $kx^2 - (3k - 1)x + k$ positive definite? 3
- f) Prove that: $\frac{\cos A}{1 + \sin A} + \frac{1 + \sin A}{\cos A} = 2 \sec A$ 3

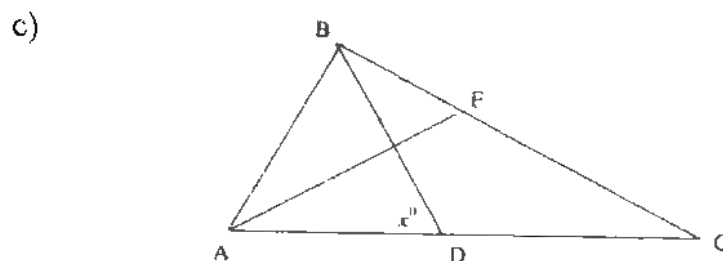
Question 2 (15 marks) *Start a new page*

- a) Differentiate with respect to x .
- i. $-5(2 - \pi x)^3$ 1
- ii. $(e^x - e^{-x})^2$ 1
- iii. $2x \log_e x$ 2
- iv. $\frac{2x + 4}{x - 1}$ 2
- b) A lighthouse stands on top of a 20 m high cliff. From a ship anchored off shore, the angle of elevation of the top of the lighthouse is 65° and the angle of elevation of the base of the lighthouse is $63^\circ 12'$. Find, to the nearest metre, the height of the top of the lighthouse above sea level. 3
- c) The distance between the point $(x, -2)$ and the line $4x - 3y + 2 = 0$ is 8 units. Find two possible values of x . 2

- d) i. Sketch the graphs of $y = \cos x$ and $y = \frac{3}{2} \tan x$ from $x = -\frac{\pi}{2}$ to $x = \frac{\pi}{2}$ on the same set of axes 1
- ii. By solving the equation $\cos x = \frac{3}{2} \tan x$, find the coordinates of the point of intersection of the two graphs that lie between $x = 0$ and $x = \frac{\pi}{2}$. 3

Question 3 (15 marks) *Start a new page*

- a) Write down the equation of the parabola with the vertex at the origin and directrix $x = 2$. 1
- b) One root of the equation $x^2 - (r + 3)x + 2(r + 1) = 0$ is twice the other root. Find the two possible values of r . 3

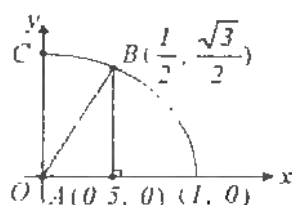


$\triangle ABC$ is a right-angled triangle with $\angle ABC = 90^\circ$. D is a point on AC such that $AB = BD = DC$. E lies on BC such that AE bisects $\angle BAD$. Let $\angle ADB = x^\circ$. Copy the diagram showing this information.

- i. Show that $\angle DBC = (2x - 90)^\circ$. 1
- ii. Hence find the value of x . 1
- iii. Show that $\triangle AEC$ is isosceles. 2
- d) The function $y = x^3 - 3x^2 - 9x + 1$ is defined in the domain $-4 \leq x \leq 5$.
- i. Find the co-ordinates of any turning points and determine their nature. 3
- ii. Find the coordinates of any points of inflexion. 2
- iii. Draw a neat sketch of the curve for the defined domain. 2

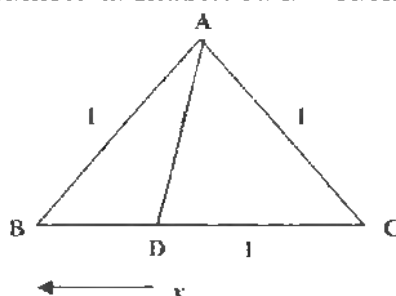
Question 4 (15 marks) *Start a new page*

- a) The diagram shows the first quadrant of the circle $x^2 + y^2 = 1$. The point A has coordinates $\left(\frac{1}{2}, 0\right)$ and $B\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$.



Not to scale

- What is the exact value of $\angle C'OB$? 1
 - Show that the exact value of the shaded area $OABC'$ is $\frac{2\pi + 3\sqrt{3}}{24}$. 2
- b) In the diagram, ABC' is an isosceles triangle where $\angle BAC' = \frac{3\pi}{5}$ and $AB = AC' = 1$. The point D is chosen on BC' such that $CD = 1$. Let $BC' = x$.



Not to scale

- Show that $\angle ADC' = \frac{2\pi}{5}$. 2
 - Given that $\triangle DBA$ and $\triangle ABC'$ are similar, deduce that $x^2 - x - 1 = 0$. 1
 - By using the cosine rule in $\triangle ABC'$, deduce that $\cos \frac{3\pi}{5} = \frac{1 + \sqrt{5}}{4}$. 3
- c) The point $P\left(n, \frac{n^2}{2}\right)$ lies on the parabola with equation $2y = x^2$.
 $A(4, 1)$ is a fixed point
- Show that $PA = \left(\frac{n^4}{4} - 8n + 17\right)^{\frac{1}{2}}$. 2
 - As P moves, find the minimum distance of P from A . 3
 - Prove that for this position of P , AP is normal to the parabola. 1

End of the paper

Question 1

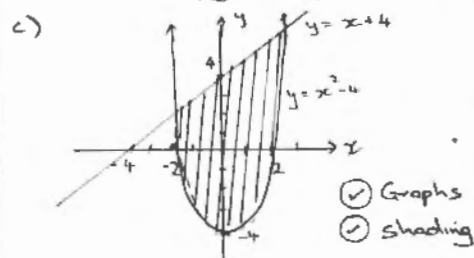
a) i) $\sin 150 = \sin 30$
 $= \frac{1}{2}$ ✓
 ii) $\operatorname{cosec} 315 = -\operatorname{cosec} 45$
 $= -\frac{1}{\sin 45}$
 $= -\sqrt{2}$ ✓

b) $3 \log_{10} 4 - 2 \log_{10} x = 6$
 $\log_{10} 4^3 - \log_{10} x^2 = 6$ ✓
 $\log_{10} \left(\frac{4^3}{x^2}\right) = 6$

$$10^6 = \frac{4^3}{x^2}$$
 ✓

$$x^2 = \frac{4^3}{10^6}$$

$$x = \frac{8}{10^3}$$
 ✓ #



d) $\frac{\sqrt{2}+1}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1}$ ✓
 $= \frac{(\sqrt{2}+1)^2}{2-1}$

$$= \frac{2+2\sqrt{2}+1}{1}$$

$$= 3+2\sqrt{2}$$
 ✓

$$\therefore p = 3 \text{ and } q = 2$$

e) $kx^2 - (3k+1)x + k$
 $b^2 - 4ac < 0$ and $k > 0$

$$\begin{cases} a = k \\ b = -(3k+1) \\ c = k \end{cases}$$

$$(-3k+1)^2 - 4(k)(k) < 0$$
 ✓

$$9k^2 - 6k + 1 - 4k^2 < 0$$

$$5k^2 - 6k + 1 < 0$$

$$(5k-1)(k-1) < 0$$
 ✓

$$\left\{ \frac{1}{5} < k < 1 \right\}$$
 ✓

f) $\frac{\cos A}{1 + \sin A} + \frac{1 + \sin A}{\cos A} = 2 \sec A$

LHS

$$= \frac{\cos^2 A + (1 + \sin A)^2}{(1 + \sin A) \cos A}$$
 ✓

$$= \frac{\cos^2 A + 1 + 2 \sin A + \sin^2 A}{(1 + \sin A) \cos A}$$

$$= \frac{\cos^2 A + \sin^2 A + 1 + 2 \sin A}{(1 + \sin A) (\cos A)}$$

$$= \frac{1 + 1 + 2 \sin A}{(1 + \sin A) (\cos A)}$$
 ✓

$$= \frac{2(1 + \sin A)}{(1 + \sin A) \cos A}$$

$$= \frac{2}{\cos A}$$

$$= 2 \sec A$$

$$= \text{RHS}$$
 ✓

Question 2

i) $-15\pi(2-\pi x)^2$ ✓

ii) $2e^{2x} - 2e^{-2x}$
 or
 $2(e^k + e^{-k})(e^k - e^{-k})$ ✓

iii) $2x \log x$
 let $u = 2x$ and $u' = 2$
 $v = \log x$ $v' = \frac{1}{x}$

$$\frac{dy}{dx} = uv' + vu'$$

$$= x + 2 \ln x$$
 ✓

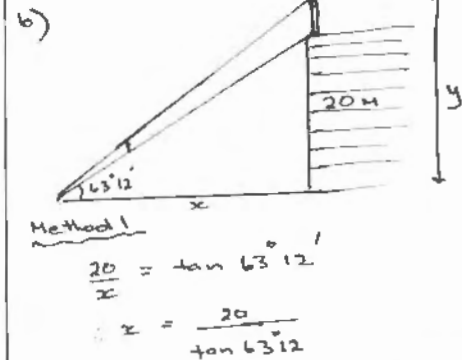
iv) $\frac{2x+4}{x-1}$
 let $u = 2x+4$ and $u' = 2$
 $v = x-1$ and $v' = 1$

Quotient Rule

$$\frac{vu' - uv'}{v^2}$$

$$= \frac{2(x-1) - (2x+4)}{(x-1)^2}$$
 ✓
$$= \frac{2x-2-2x-4}{(x-1)^2}$$

$$= \frac{-6}{(x-1)^2}$$
 ✓



$$\frac{y}{x} = \tan 65^\circ$$

$$y = x \tan 65^\circ$$

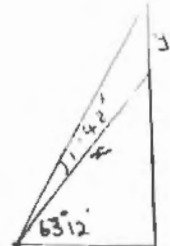
$$= \frac{20 \tan 65^\circ}{\tan 63.12^\circ}$$

$$y = 22m$$

Method 2

$$\sin 63.12^\circ = \frac{20}{x}$$

$$x = \frac{20}{\sin 63.12^\circ}$$



Using sine rule

$$\frac{y}{\sin 48^\circ} = \frac{20}{\sin 63.12^\circ \sin 25^\circ}$$

$$= 1.6$$

$$\therefore \text{Total height} = 20 + 1.6$$

$$= 21.6$$

$$= 22m$$

c) $(x, -2)$ and $4x - 3y + 2 = 0$

$$d = \frac{|ax + by + c|}{\sqrt{a^2 + b^2}}$$

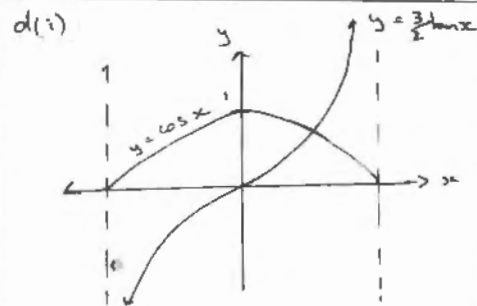
$$8 = \frac{|4x + 6 + 2|}{\sqrt{3^2 + 4^2}}$$
 ✓

$$8 = \frac{|4x + 8|}{5}$$

$$14x + 81 = 40$$

$$4x + 8 = 40 \quad -4x - 8 = 40$$

$$x = 8 \quad x = -12$$
 ✓



Graph

(ii) $\cos x = \frac{3}{2} \tan x$
 $2 \cos x = 3 \frac{\sin x}{\cos x}$
 $2 \cos^2 x = 3 \sin x$
 $2(1 - \sin^2 x) = 3 \sin x$
 $2 - 2 \sin^2 x = 3 \sin x$
 $0 = 2 \sin^2 x + 3 \sin x - 2$

let $x = \sin x$

$0 = 2x^2 + 3x - 2$
 $0 = (2x - 1)(x + 2)$

$x = \frac{1}{2}$ or $x = -2$

$\sin x = \frac{1}{2}$ or $\sin x = -2$

$x = 30^\circ$
 $= \frac{\pi}{6}$

No solution

$\therefore$ pt of intersection is $(\frac{\pi}{6}, \frac{\sqrt{3}}{2})$

Question 3

a) $y^2 = -8x$



b) $(x+2)^2 = 3x = \frac{1}{2}x + 3$

$\therefore x = \frac{r+3}{3}$

$2x = 2x^2 = \frac{r}{2} = 2r+2$

$x = \pm \sqrt{r+1}$

Solve simultaneously

$\frac{r+3}{3} = \sqrt{r+1}$

$r+3 = 3\sqrt{r+1}$

$(r+3)^2 = 9(r+1)$

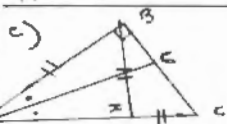
$r^2 + 6r + 9 = 9r + 9$

$r^2 - 3r = 0$

$r(r-3) = 0$

$r = 0$ or $r = 3$

$\therefore r = 0$ or 3



i) $\angle ADB = \angle BAD = x$ (equal sides of isosceles triangle)

$\angle ABD = 180 - 2x$ (angle sum of $\triangle$)

$\therefore \angle OBC = 90 - (180 - 2x)$

$= 2x - 90$

ii) $x = 60^\circ$

iii) $\angle BAD = 60^\circ$

$\therefore \angle EAD = 30^\circ$ — from part (i)

$\angle OBC = \angle DCB$ (equal angles opposite equal sides in $\triangle ABC$)

$= 2x - 90$

$= 2(60) - 90$ from part (ii)

$= 30^\circ$

$\therefore \angle EAC$ and $\angle DCB$ are both

30° — form the base angles

of an isosceles $\triangle$.

d) $y = x^3 - 3x^2 - 9x + 1$

i) TP $\frac{dy}{dx} = 0$

$3x^2 - 6x - 9 = 0$

$3(x^2 - 2x - 3) = 0$

$3(x-3)(x+1) = 0$

$x = 3$ or $x = -1$

when $x = 3 \Rightarrow y = -26$

when $x = -1 \Rightarrow y = 6$

$\therefore$ TP occur at $(3, -26)$ and $(-1, 6)$

Nature

at $x = 3$

x	2	3	4
$\frac{dy}{dx}$	-3	0	5
	(+ve)	-	(-ve)

$\frac{d^2y}{dx^2} = 12$
 > 0

$\therefore$ relative minimum point

at $x = -1$

x	-2	-1	0
$\frac{dy}{dx}$	5	0	-3
	(+ve)	-	(-ve)

$\frac{d^2y}{dx^2} = -12$
 < 0

$\therefore$ relative maximum point

ii) Point of inflexion

$\frac{d^2y}{dx^2} = 0$

$6x - 6 = 0$

$x = 1$

at $x = 1$ $y = -10$

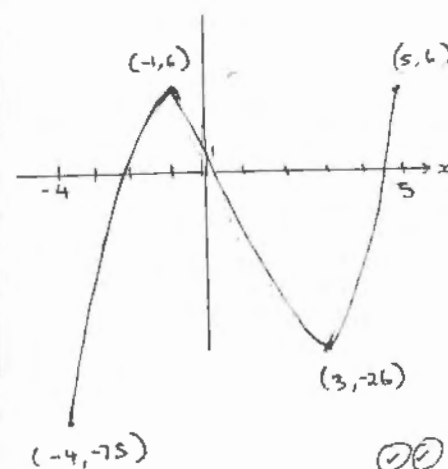
Test

x	0	1	2
$\frac{d^2y}{dx^2}$	-6	-	+6

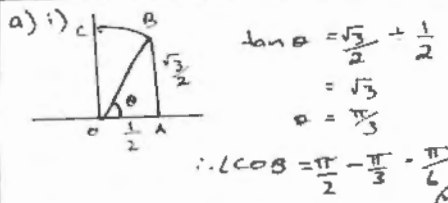
Concavity changes

point of inflexion occurs at $(1, -10)$

iii)



Question 4



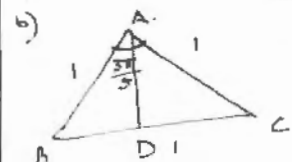
ii) Total Area = Area of $\triangle OAB$ + Area of $\triangle ABC$

$= \frac{1}{2} r^2 \theta + \frac{1}{2} bh$

$= \frac{1}{2} (1^2) \cdot \frac{\pi}{6} + \frac{1}{2} \times \frac{1}{2} \times \frac{\sqrt{3}}{2}$

$= \frac{\pi}{12} + \frac{\sqrt{3}}{8}$

$= \frac{2\pi + 3\sqrt{3}}{24}$



$\angle ACB = (\pi - \frac{3\pi}{5}) \div 2$
 $= \frac{2\pi}{5} \div 2$
 $= \frac{\pi}{5}$

(Equal angles opposite equal sides in $\triangle ABC$)

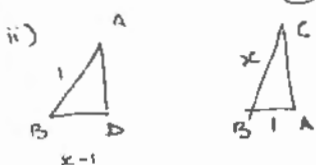
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$\angle CAD = \angle CDA$ (Equal angles are opposite equal sides in $\triangle ACD$)
 $= (\pi - \frac{\pi}{5}) \div 2$

$= \frac{4\pi}{5} \div 2$

$= \frac{4\pi}{10}$

$= \frac{2\pi}{5}$



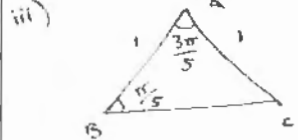
ratio of similar triangles

$\frac{x}{1} = \frac{1}{x-1}$

$x(x-1) = 1$

$x^2 - x = 1$

$x^2 - x - 1 = 0$



$x^2 - x - 1 = 0$
 $a=1, b=-1, c=-1$

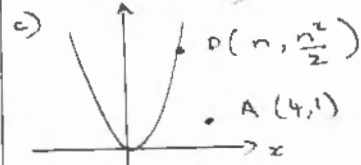
$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
 $= \frac{1 \pm \sqrt{5}}{2}$

Length is $\frac{1+\sqrt{5}}{2}$

$\cos \frac{\pi}{5} = \frac{(\frac{1+\sqrt{5}}{2})^2 + (1)^2 - 1^2}{2(\frac{1+\sqrt{5}}{2}) \cdot 1}$

$= \frac{(\frac{1+\sqrt{5}}{2})^2}{(1+\sqrt{5})}$

$= \frac{(1+\sqrt{5})(1+\sqrt{5})}{4(1+\sqrt{5})}$
 $= \frac{1+\sqrt{5}}{4}$



$PA = \sqrt{(4-n)^2 + (1 - \frac{n^2}{2})^2}$

$= \sqrt{16 - 8n + n^2 + 1 - n^2 + \frac{n^4}{4}}$

$= \sqrt{\frac{n^4}{4} - 8n + 17}$

$= (\frac{n^4}{4} - 8n + 17)^{\frac{1}{2}}$

ii) Minimum distance occurs when $\frac{d(PA)}{dn} = 0$

$0 = \frac{1}{2}(\frac{n^4}{4} - 8n + 17)^{-\frac{1}{2}} \times (n^3 - 8)$

$0 = \frac{n^3 - 8}{2\sqrt{\frac{n^4}{4} - 8n + 17}}$

$\therefore n^3 - 8 = 0$
 $n^3 = 8$
 $n = \sqrt[3]{8}$
 $n = 2$

when $n=2$

$PA = (\frac{2^4}{4} - 8(2) + 17)^{\frac{1}{2}}$
 $= \sqrt{5}$ units

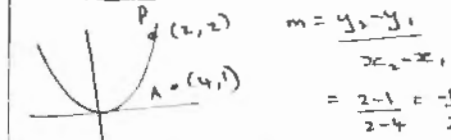
Test Minimum

x	1	2	3
$\frac{d(PA)}{dn}$	-7	0	15
$\frac{d^2(PA)}{dn^2}$	-14	0	30
	-ve	0	+ve

This is a local minimum and with only one TP is also absolute minimum.

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ii) gradient of PA



$m = \frac{y_2 - y_1}{x_2 - x_1}$
 $= \frac{2 - 1}{2 - 4} = -\frac{1}{2}$

equation of tangent of parabola at $x=2$

$y = \frac{1}{2}x^2$
 $y' = x$
 at $x=2$ gradient is 2.

$\therefore m_1 \cdot m_2 = -1$
 $2 \times -\frac{1}{2} = -1$
 true

$\therefore PA$ is $\perp$ to tangent of parabola at P